

RHCE 9

1. Install and Configure Ansible on the control node as follows:

- * Install the required packages.
- * Create a static inventory file called /home/student/ansible/inventory as follows:
 - node1 is a member of the dev host group
 - node2 is a member of the test host group
 - node3 and node4 is a member of the prod host group
 - node5 is a member of the balancers host group
 - The prod group is a member of the webserver host group
- * Create a configuration file called ansible.cfg as follows:
 - The host inventory file /home/student/ansible/inventory is defined
 - The location of roles used in playbooks is defined as /home/student/ansible/roles
 - The Install Ansible Content Collection location /home/student/ansible/mycollection

Solution --

```
# sudo yum install ansible-navigator ansible* tree vim -y (In Exam it will work)
```

```
# mkdir /home/student/ansible
```

```
# mkdir /home/student/ansible/roles
```

```
# mkdir /home/student/ansible/mycollection
```

```
# cd /home/student/ansible
```

```
# vim inventory
```

```
a
```



```
[dev]
servera
[test]
serverb
[prod]
serverc
serverd
[balancers]
servere
[webserver:children]
prod
```

```
# vim ansible.cfg
```

```
[defaults]
```

```
remote_user=devops
```

```
inventory=/home/student/ansible/inventory
```

```
roles_path=/home/student/ansible/roles
```

```
collection_path=/home/student/ansible/mycollection:~/.ansible/collections:/usr/share/ansible/
collections
```

```
[privilege_escalation]
```

```
become=true
```

```
become_method=sudo
become_user=root
```

To check all is working

```
# ansible all -m ping
```

2. Create repo.yml for configuring repository in all nodes.

i) Name = baseos-internal
Description = Baseos Description
Url = http://content/rhel9.0/x86_64/dvd/:set
GPG is enabled.
Gpgkey = http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
Repository is enabled.

ii) Name = appstream-internal
Description = App Description
Url = http://content/rhel9.0/x86_64/dvd/AppStream
GPG is enabled.
Gpgkey = http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
Repository is enabled.

Solution --

```
---
- name: configure repository
  hosts: all
  tasks:
    - name: Import a key from a url
      ansible.builtin.rpm_key:
        state: present
        key: http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release

    - name: Add multiple repositories into the same file (1/2)
      ansible.builtin.yum_repository:
        name: AppStream
        description: AppStream
        file: appstream
        baseurl: https://download.fedoraproject.org/pub/epel/$releasever/$basearch
        gpgcheck: yes
        gpgkey: http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
        enabled: yes

    - name: Add multiple repositories into the same file (1/2)
      ansible.builtin.yum_repository:
        name: BaseOS
        description: BASEos
        file: Baseos
        baseurl: https://download.fedoraproject.org/pub/epel/$releasever/$basearch
        gpgcheck: yes
        gpgkey: http://content.example.com/rhel9.0/x86_64/dvd/RPM-GPG-KEY-redhat-release
        enabled: yes
```

Note: [Go inside any manage node and install ftp package for test purpose] repository working or not

```
# yum install ftp
```

3. Create a directory 'roles' under /home/student/ansible .

i) Create a playbook called requirements.yml under the roles directory and download the given roles under the 'roles' directory using galaxy command under it.

ii) Role name should be balancer and download using this url

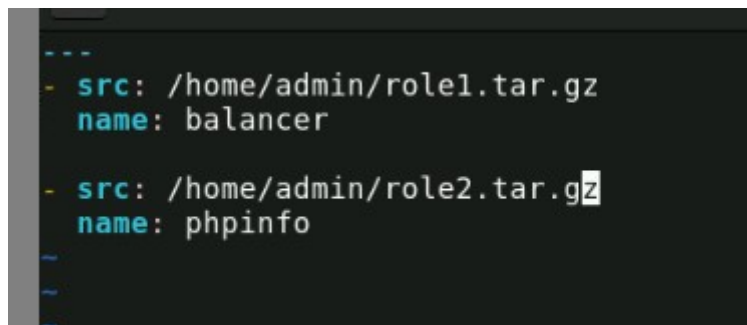
<http://content.example.com/Rhce/balancer.tgz>.

iii) Role name phpinfo and download using this url <http://content.example.com/Rhce/phpinfo.tgz>.

Solution --

```
# cd /home/student/ansible/roles
```

```
# vim requirements.yml
```



```
-- --
- src: /home/admin/role1.tar.gz
  name: balancer

- src: /home/admin/role2.tar.gz
  name: phpinfo
```

run this --

```
ansible-galaxy install -r requirements.yml -p .
```

Note: [check 2 dir created one is name balancer and 2nd is name phpinfo]

4. Create offline role named apache under roles directory.

i) Install httpd package and the service should be start and enable the httpd service.

ii) Host the web page using the template.j2

iii) The template.j2 should contain

Welcome to HOSTNAME ON IPADDRESS

Where HOSTNAME is fully qualified domain name.

iv) Create a playbook named apache_role.yml and run the role in webservers group.

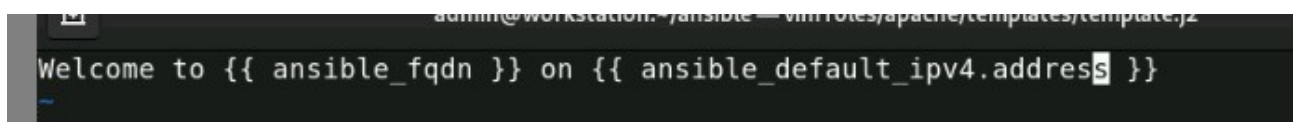
Solution --

```
# cd /home/student/ansible/roles
```

```
# ansible-galaxy init apache
```

```
# cd apache/
```

```
# vim templates/template.j2
```



```
admin@workstation: ~/ansible -- vim roles/apache/templates/template.j2
Welcome to {{ ansible_fqdn }} on {{ ansible_default_ipv4.address }}
```

vim tasks/main.yml

```
- name: Install the latest version of Apache
  ansible.builtin.dnf:
    name:
      - httpd
    state: latest

- name: Start service httpd, if not started
  ansible.builtin.service:
    name: httpd
    state: started
    enabled: yes

- name: Start service firewalld
  ansible.builtin.service:
    name: firewalld
    state: restarted
    enabled: yes

- name: Add http service in firewall rule
  ansible.builtin.firewalld:
    service: http
    state: enabled
    permanent: yes
    immediate: yes

- name: Copy the template.j2 file to web server directory
  ansible.builtin.template:
    src: template.j2
    dest: /var/www/html/index.html
```

vim /home/student/ansible/apache_role.yml

```
---
- name: use apache role
  hosts: webservers
  roles:
    - apache
```

ansible-playbook apache_role.yml

```
[admin@workstation ansible]$ curl serverc
Welcome to serverc.lab.example.com on 172.25.250.12
[admin@workstation ansible]$ curl serverd
Welcome to serverd.lab.example.com on 172.25.250.13
[admin@workstation ansible]$
```

6. Installing Ansible Content Collections

<http://rhls.lab.example.com/materials>
tar file

installs the collection in the local collections directory.
download tar file url given on exam

```
$ cd /home/student/ansible/mycollection
```

collections:

```
$ wget http://server.lab.example.com/role-collections/redhat-insights-1.0.7.tar.gz
$ wget http://server.lab.example.com/role-collections/redhat-rhel_system_roles-1.19.3.tar.gz
$ wget http://server.lab.example.com/role-collections/community-general-5.5.0.tar.gz
```

```
[admin@workstation mycollection]$ ll
total 3012
-rw-r--r--. 1 admin admin 2235450 Jul 12 15:42 community-general-5.5.0.tar.gz
-rw-r--r--. 1 admin admin   34218 Jul 12 15:42 redhat-insights-1.0.7.tar.gz
-rw-r--r--. 1 admin admin  808333 Jul 12 15:42 redhat-rhel_system_roles-1.19.3.tar.gz
[admin@workstation mycollection]$
```

```
$ ansible-galaxy collection install /home/student/ansible/mycollection/redhat-insights-1.0.7.tar.gz
```

```
$ ansible-galaxy collection install /home/student/ansible/mycollection/redhat-rhel_system_roles-1.19.3.tar.gz
```

```
$ ansible-galaxy collection install /home/student/ansible/mycollection/community-general-5.5.0.tar.gz
```

```
-rw-r--r--. 1 admin admin   34218 Jul 12 15:42 redhat-insights-1.0.7.tar.gz
-rw-r--r--. 1 admin admin  808333 Jul 12 15:42 redhat-rhel_system_roles-1.19.3.tar.gz
[admin@workstation mycollection]$ ansible-galaxy collection install community-general-5.5.0
.tar.gz
Starting galaxy collection install process
Process install dependency map
Starting collection install process
Installing 'community.general:5.5.0' to '/home/admin/.ansible/collections/ansible_collections/community/general'
community.general:5.5.0 was installed successfully
[admin@workstation mycollection]$ ansible-galaxy collection install redhat-insights-1.0.7.t
ar.gz
Starting galaxy collection install process
Process install dependency map
Starting collection install process
Installing 'redhat.insights:1.0.7' to '/home/admin/.ansible/collections/ansible_collections/redhat/insights'
redhat.insights:1.0.7 was installed successfully
[admin@workstation mycollection]$ ansible-galaxy collection install redhat-rhel_system_role
s-1.19.3.tar.gz
Starting galaxy collection install process
Process install dependency map
Starting collection install process
Installing 'redhat.rhel_system_roles:1.19.3' to '/home/admin/.ansible/collections/ansible_collections/redhat/rhel_system_roles'
redhat.rhel_system_roles:1.19.3 was installed successfully
[admin@workstation mycollection]$
```

on safer side

```
$ansible-galaxy collection install /home/student/ansible/mycollection/redhat-insights-1.0.7.tar.gz -p /home/student/ansible/mycollection
```

```
$ansible-galaxy collection install /home/student/ansible/mycollection/redhat-rhel_system_roles-1.19.3.tar.gz -p /home/student/ansible/mycollection
```

```
$ansible-galaxy collection install /home/student/ansible/mycollection/community-general-5.5.0.tar.gz -p /home/student/ansible/mycollection
```

7. Install packages in multiple group.

- i) Install php and mariadb-server packages in dev and test group.
- ii) Install "RPM Development Tools" group package in prod group.
- iii) Update all packages in dev group.
- iv) Use separate play for each task and playbook name should be packages.yml.

vim packages.yml

```
---
- name: package installation
  hosts: dev,test,prod
  tasks:
    - name: installing php and mariadb-server
      ansible.builtin.dnf:
        name:
          - php
          - mariadb-server
        state: present
- name: group package installation
  hosts: prod
  tasks:
    - name: installing group package 'Development tools'
      ansible.builtin.dnf:
        name: '@RPM Development Tools'
        state: present
- name: update packages
  hosts: dev
  tasks:
    - name: updating all
      ansible.builtin.dnf:
        name: '*'
        state: latest
```

8. Create a playbook webcontent.yml and it should run on dev group.

- i) Create a directory /devweb and it should be owned by devops group.
- ii) /devweb directory should have context type as "httpd"
- iii) Assign the permission for user=rwx,group=rwx,others=rx and group special permission should be applied to /devweb.
- iv) Create an index.html file under /devweb directory and the file should have the content "Development".
- v) Link the /devweb directory to /var/www/html/devweb.

Vim webcontent.yml

ansible-playbook webcontent.yml

```
---
- name: create web content dir
  hosts: dev
  tasks:
    - name: create a dir
      ansible.builtin.file:
        path: /devweb
        state: directory
        mode: '2755'
        group: devops
        setype: httpd_sys_content_t

    - name: Create a symbolic link
      ansible.builtin.file:
        src: /devweb
        dest: /var/www/html/devweb
        state: link
        setype: httpd_sys_content_t

    - name: Copy using inline content
      ansible.builtin.copy:
        content: "Development"
        dest: /devweb/index.html
        setype: httpd_sys_content_t

    - name: permit traffic in default zone for https service
      ansible.posix.firewalld:
        service: http
        permanent: true
        state: enabled
        immediate: true
```

curl <http://node1.lab.example.com/devweb/>

```
[admin@workstation ansible]$ curl servera/devweb/
Development[admin@workstation ansible]$
```

10. Replace the file /etc/issue on all managed nodes.

- i) In dev group /etc/issue should have the content "Developement".
- ii) In test group /etc/issue should have the content "Test".
- iii) In prod group /etc/issue should have the content "Production".
- iv) Playbook name issue.yml and run in all managed nodes.

Solution --

vim issue.yml

```
--
- name: content in dev
  hosts: dev
  tasks:
    - name: Copy using inline content
      ansible.builtin.copy:
        content: Development
        dest: /etc/issue

- name: content in dev
  hosts: prod
  tasks:
    - name: Copy using inline content
      ansible.builtin.copy:
        content: Production
        dest: /etc/issue

- name: content in dev
  hosts: test
  tasks:
    - name: Copy using inline content
      ansible.builtin.copy:
        content: Test
        dest: /etc/issue
```

result

```
[admin@workstation ansible]$ ansible -m command -a "cat /etc/issue" all
servera | CHANGED | rc=0 >>
Development
serverc | CHANGED | rc=0 >>
Production
serverd | CHANGED | rc=0 >>
Production
serverb | CHANGED | rc=0 >>
Test
servere | CHANGED | rc=0 >>
\S
Kernel \r on an \m
[admin@workstation ansible]$
```

11. Download the file <http://content.example.com/Rhce/myhosts.j2>.

i) myhosts.j2 is having the content.

```
127.0.0.1 localhost.localdomain localhost
192.168.0.1 localhost.localdomain localhost
```

ii) The file should collect all node information like ipaddress,fqdn,hostname and it should be the same as in the /etc/hosts file, if playbook run in all the managed node it must store in /etc/myhosts.

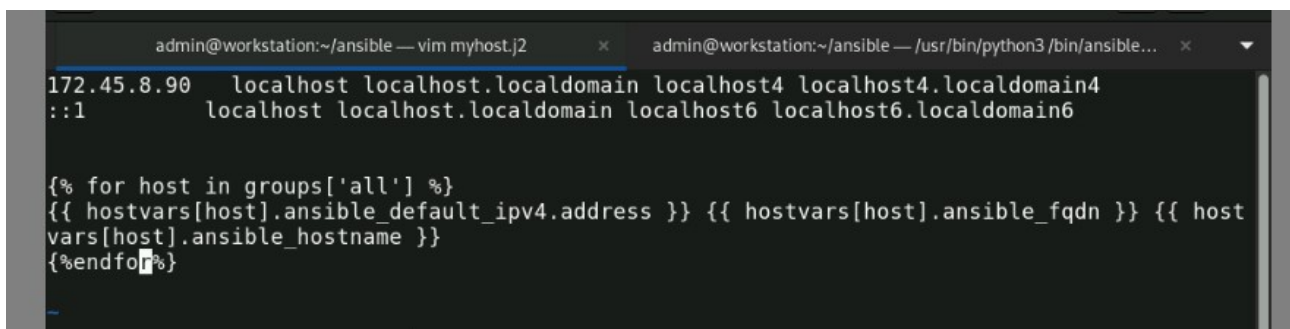
Finally /etc/myhosts file should contains like.

```
127.0.0.1 localhost.localdomain localhost
192.168.0.1 localhost.localdomain localhost

172.25.250.10 node1.lab.example.com node1
172.25.250.11 node2.lab.example.com node2
172.25.250.12 node3.lab.example.com node3
172.25.250.13 node4.lab.example.com node4
172.25.250.14 node4.lab.example.com node5
```

iii) playbook name hosts.yml and run in dev group.

Vim myhost.j2



```
admin@workstation:~/ansible — vim myhost.j2  admin@workstation:~/ansible — /usr/bin/python3 /bin/ansible...
172.45.8.90  localhost localhost.localdomain localhost4 localhost4.localdomain4
::1        localhost localhost.localdomain localhost6 localhost6.localdomain6

{% for host in groups['all'] %}
{{ hostvars[host].ansible_default_ipv4.address }} {{ hostvars[host].ansible_fqdn }} {{ host
vars[host].ansible_hostname }}
{%endfor%}
```

```
{% for host in groups['all'] %}
{{ hostvars[host].ansible_default_ipv4.address }}
{{ hostvars[host].ansible_fqdn }} {{ hostvars[host].ansible_hostname }}
{%endfor%}
```

we have to write this in file

```
admin@workstation:~/ansible — vim host.yml  x  admin@workstation:~/ansible
--
- name: use template
  hosts: all
  tasks:
    - name: use template
      ansible.builtin.template:
        src: ./myhost.j2
        dest: /etc/myhosts

- name: delet from all
  hosts: all , !dev
  tasks:
    - name: delete file
      ansible.builtin.file:
        path: /etc/myhosts
        state: absent
```

to check

```
[admin@workstation ansible]$ ansible -m command -a "cat /etc/myhosts" all
serverc | FAILED | rc=1 >>
cat: /etc/myhosts: No such file or directorynon-zero return code
servera | CHANGED | rc=0 >>
172.45.8.90 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost localhost.localdomain localhost6 localhost6.localdomain6

172.25.250.10 servera.lab.example.com servera
172.25.250.11 serverb.lab.example.com serverb
172.25.250.220 serverc.lab.example.com serverc
172.25.250.12 serverd.lab.example.com serverd
servere | FAILED | rc=1 >>
cat: /etc/myhosts: No such file or directorynon-zero return code
serverd | FAILED | rc=1 >>
cat: /etc/myhosts: No such file or directorynon-zero return code
serverb | FAILED | rc=1 >>
cat: /etc/myhosts: No such file or directorynon-zero return code
[admin@workstation ansible]$
```

5. Create a playbook called roles.yml and it should run balancer and phpinfo roles.

Use roles from Ansible Galaxy

Create a playbook called /home/admin/ansible/roles.yml

* The palybook contains a play that runs on host in the balances host group and uses the balances role.

- This role configures a service to load balance web server request between hosts in the webserver host group.

- Browsing to host in the balances host group (for example <http://node5.lab.example.com>) produces the following output:

Welcome to node3.lab.example.com on 172.25.250.12

- Reloading the Browser produces output from the alternate web server:

Welcome to node4.lab.example.com on 172.25.250.13

* The Playbook contains a play that runs on hosts in the webserver host group and uses the phpinfo role.

- Browsing to host in the webserver host group with the URL /hell.php produces the following output:

Hello PHP World from FQDN

- For example, browsing to <http://node3.lab.example.com/hello.php> produces the following output:

Hello PHP World from node3.lab.example.com

along with various details of the PHP configuration including the version of PHP that is installed.

- Similarly, browsing to <http://node4.lab.example.com/hello.php>, produces the following output:

Hello PHP World from node4.lab.example.com

along with various details of the PHP configuration including the version of PHP that is installed

Solution ---

vim roles/balancer/template/balancer.j2

```
#-----
# main frontend which proxies to the backends
#-----
frontend serverc.lab.example.com
  bind *:80
    acl url_static      path_beg      -i /static /images /javascript /stylesheets
    acl url_static      path_end      -i .jpg .gif .png .css .js

    use_backend static   if url_static
    default_backend      app

#-----
# static backend for serving up images, stylesheets and such
#-----
backend static
  balance roundrobin
  server static 127.0.0.1:4331 check

#-----
# round robin balancing between the various backends
#-----
backend app
  balance roundrobin
  server serverc.lab.example.com 172.25.250.12:80 check
  server serverd.lab.example.com 172.25.250.13:80 check
```

cd /home/admin/ansible

vim balancer.yml

and use both the roles on playbooks

```
---
- name: use balancer role
  hosts: balancers
  roles:
    - balancer

- name: use phpinfo
  hosts: webserver
  roles:
    - phpinfo
~
~
```

curl servere

```
[admin@workstation ansible]$ curl servere
Welcome to serverc.lab.example.com on 172.25.250.12
[admin@workstation ansible]$ curl servere
Welcome to serverd.lab.example.com on 172.25.250.13
[admin@workstation ansible]$ curl servere
Welcome to serverc.lab.example.com on 172.25.250.12
[admin@workstation ansible]$ curl servere
Welcome to serverd.lab.example.com on 172.25.250.13
[admin@workstation ansible]$
```

14. Rekey the variable file from <http://content.example.com/Rhce/solaris.yml>.

- i) Old password: cisco
- ii) New password: redhat

Solution --

```
# wget http://content/Rhce/solaris.yml
```

```
# ansible-vault rekey solaris.yml
```

Old password:

New password:

Confirm new password:

```
[admin@workstation ansible]$ ansible-vault rekey solaris.yml
Vault password:
New Vault password:
Confirm New Vault password:
Rekey successful
[admin@workstation ansible]$ █
```

12. Create a variable file vault.yml and that file should contains the variable and its value.

pw_developer is value lamdev

pw_manager is value lammgr

- i) vault.yml file should be encrypted using the password "P@sswOrd".
- ii) Store the password in secret.txt file and which is used for encrypt the variable file.

Solution --

```
# vim secret.txt
P@sswOrd
```

```
# ansible-vault create vault.yml --vault-password-file=secret.txt
```

```
pw_developer: lamdev
pw_manager: lammgr
```

```
-----
[admin@workstation ansible]$ vim secret.yml
[admin@workstation ansible]$ ansible-vault create vault.yml --vault-password-file secret.yml
[admin@workstation ansible]$ ansible-vault view vault.yml --vault-password-file secret.yml
pw_developer: lamdev
pw_manager: lammgr
[admin@workstation ansible]$ █
```

put entry in ansible.cfg file
vault_password_file=./secrete.yml

```
[defaults]
inventory=/home/admin/ansible/inventory
remote_user=admin
roles_path=/home/admin/ansible/roles
collection_path=/home/admin/ansible/mycollection:~/.ansible/collections:/usr/share/ansible/collections
vault_password_file=./secret.yml

[privilege_escalation]
become=true
become_method=sudo
become_user=root
```

13. Download the variable file http://content.example.com/Rhce/user_list.yml and
Playbook name users.yml and run in all nodes using two variable files user_list.yml and
vault.yml

- i) * Create a group opsdev
 - * Create user from users variable who's job is equal to developer and need to be in opsdev group
 - * Assign a password using SHA512 format and run the playbook on dev and test.
 - * User password is {{ pw_developer }}
- ii) * Create a group opsmgr
 - * Create user from users variable who's job is equal to manager and need to be in opsmgr group
 - * Assign a password using SHA512 format and run the playbook on prod.
 - * User password is {{ pw_manager }}
- iii)* Use when condition for each play.

Solution --

wget http://content.example.com/Rhce/user_list.yml

we get the user_list.yml

```

---
- user:
    - name: joe
      job: developer

    - name: bob
      job: manager

    - name: alice
      job: manager
~
~

```

vim hosts.yml

```

hosts: dev,test
vars_files:
  - user_list.yml
  - vault.yml
tasks:
  - name: add group
    group:
      name: devops
      state: present

  - name: adding users
    user:
      name: "{{ item.name }}"
      groups: devops
      password: "{{ pw_developer | password_hash('sha512') }}"
    when: item.job=='developer'
    loop: "{{ user }}"

- name: Create an users and groups
  hosts: prod
  vars_files:
    - user_list.yml
    - vault.yml
  tasks:
    - name: add group
      group:
        name: opsmgr
        state: present

    - name: adding users
      user:
        name: "{{ item.name }}"
        groups: opsmgr
        password: "{{ pw_developer | password_hash('sha512') }}"
      when: item.job=='manager'
      loop: "{{ user }}"

```

check opsmgr & devops group in all nodes

15. Create a cronjob for user student in all nodes, the playbook name crontab.yml and the job details are below

i) Every 2 minutes the job will execute logger "EX294 in progress"

```
---
- name: use crontab
  hosts: all
  tasks:
    - name: use cron job
      ansible.builtin.cron:
        name: "test"
        minute: "*/2"
        user: admin
        job: logger "RH294 in progress"
        state: present
```

tho check --

```
[admin@workstation ansible]$ vim crón.yml
[admin@workstation ansible]$ ansible -m command -a "crontab -l -u admin " all
serverb | CHANGED | rc=0 >>
#Ansible: test
*/2 * * * * logger "RH294 in progress"
serverc | CHANGED | rc=0 >>
#Ansible: test
*/2 * * * * logger "RH294 in progress"
servera | CHANGED | rc=0 >>
#Ansible: test
*/2 * * * * logger "RH294 in progress"
serverd | CHANGED | rc=0 >>
#Ansible: test
*/2 * * * * logger "RH294 in progress"
servere | CHANGED | rc=0 >>
#Ansible: test
*/2 * * * * logger "RH294 in progress"
[admin@workstation ansible]$
```

17. Use a RHEL timesync system role

create a playbook called /home/admin/ansible/timesync.yml that:

- Runs on all managed nodes
- Uses the timesync role
- Configures the role to use the currently active NTP provider
- Configure the role to use the time server classroom.lab.example.com
- Configure the role to enable the iburst parameter

Solution --

vim timesync.yml

```
---
- name: use timesync
  hosts: all
  vars:
    timesync_ntp_servers:
      - hostname: marla.ludost.net
        iburst: yes

  roles:
    - rhel-system-roles.timesync

~
~
```

this file found in
/usr/share/ansible/roles

```
[admin@workstation ansible]$ ansible -m command -a "chronyc sources" all
servera | CHANGED | rc=0 >>
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* 172.25.254.254            2   8   377   209   -13us[ -91us] +/- 4006us
serverd | CHANGED | rc=0 >>
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* marla.ludost.net         2   6   377    61   -100us[ -214us] +/- 137ms
serverc | CHANGED | rc=0 >>
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* marla.ludost.net         2   6   377    62   +793us[+1112us] +/- 140ms
servere | CHANGED | rc=0 >>
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* marla.ludost.net         2   6   377   192    +40us[ +26us] +/- 136ms
serverb | CHANGED | rc=0 >>
MS Name/IP address         Stratum Poll Reach LastRx Last sample
=====
^* marla.ludost.net         2   6   377    61   -102us[ -303us] +/- 137ms
[admin@workstation ansible]$
```

17. Use Selinux role for all nodes
create selinux.yml and set selinux enforcing

```
---
- name: configure selinux
  hosts: all
  vars:
    selinux_state: enforcing
  roles:
    - role: rhel-system-roles.selinux
      become: true

~
```

ansible -m command -a "cat /etc/selinux/config" all

```
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
# See also:
# https://docs.fedoraproject.org/en-US/quick-docs/getting-started-with-selinux/#getting-started-with-selinux-selinux-states-and-modes
#
# NOTE: In earlier Fedora kernel builds, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#   grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#   grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
[admin@workstation ansible]$ []
```

9. Create hardware report using playbook in all nodes.

i) Download hwreport.txt from the url <http://content.example.com/Rhce/hwreport.empty> and save as it under /root/hwreport.txt

/root/hwreport.empty should have the content with control node informations as key=value.

```
#hwreport
HOSTNAME=
MEMORY=
BIOS=
CPU=
DISK_SIZE_VDA=
DISK_SIZE_VDB=
```

ii) If there is no information it have to show "NONE".

iii) playbook name should be hwreport.yml.

wget <http://content.example.com/Rhce/hwreport.empty>

ls

hwreport.empty

```
Inventory Hostname=hostname
Bios Version=biosversion
Total Memory=memory
VDA_SIZE=vdasize
VDB_SIZE=vdbsize
~
```

vim hwreport.yml

```

---
- name: create hwreport
  hosts: all
  tasks:
    #- name: download hwreport
    # ansible.builtin.get_url:
    #   url: http://content.example.com/Rhce/hwreport.empty
    #   dest: /root/hwreport.txt

    - name: copy from folder (only for practice)
      ansible.builtin.copy:
        src: ./hwreport.empty
        dest: /root/hwreport.txt

    - name: use replace module
      ansible.builtin.replace:
        path: /root/hwreport.txt
        regexp: "{{ item.oldvalue }}"
        replace: "{{ item.newvalue }}"
      loop:
        - oldvalue: hostname
          newvalue: "{{ ansible_hostname | default('NONE') }}"
        - oldvalue: biosversion
          newvalue: "{{ ansible_bios_version | default('NONE') }}"
        - oldvalue: memory
          newvalue: "{{ ansible_memtotal_mb | default('NONE') }}"
        - oldvalue: vdasize
          newvalue: "{{ ansible_devices.vda.size | default('NONE') }}"
        - oldvalue: vdbsize
          newvalue: "{{ ansible_devices.vdb.size | default('NONE') }}"

```

ansible -m command -a "cat /root/hwreport.txt" all

```

[admin@workstation ansible]$ ansible -m command -a "cat /root/hwreport.txt" all
serverb | CHANGED | rc=0 >>
Inventory_Hostname=serverb
Bios_Version=1.14.0-1.module+el8.4.0+8855+a9e237a9
Total_Memory=960
VDA_SIZE=10.00 GB
VDB_SIZE=1.00 GB
servera | CHANGED | rc=0 >>
Inventory_Hostname=servera
Bios_Version=1.14.0-1.module+el8.4.0+8855+a9e237a9
Total_Memory=960
VDA_SIZE=10.00 GB
VDB_SIZE=1.00 GB
serverd | CHANGED | rc=0 >>
Inventory_Hostname=serverd
Bios_Version=1.14.0-1.module+el8.4.0+8855+a9e237a9
Total_Memory=960
VDA_SIZE=10.00 GB
VDB_SIZE=1.00 GB
serverc | CHANGED | rc=0 >>
Inventory_Hostname=serverc
Bios_Version=1.14.0-1.module+el8.4.0+8855+a9e237a9
Total_Memory=960
VDA_SIZE=10.00 GB
VDB_SIZE=1.00 GB
servere | CHANGED | rc=0 >>
Inventory_Hostname=servere
Bios_Version=1.14.0-1.module+el8.4.0+8855+a9e237a9
Total_Memory=3723
VDA_SIZE=10.00 GB
VDB_SIZE=NONE
[admin@workstation ansible]$

```

16. Create & use a logical Volume

Create a playbook called /home/admin/ansible/ansible/lv.yml that runs on all managed nodes that does the following:

- * Creates a logical volume with these requirement:

- The logical Volume is Created in the research volume group
- The logical volume name is data
- The logical volume size is 1500 Mib

- * Format the logical volume with the ext4 file system

- * if the requested logical volume size cannot be created, the error message Could not create logical volume of that size should be displayed and size 800 MiB should be used instead.

- * if the volume research does not exist, the error message volume group does not exist should be displayed

- * Does NOT mount the logical volume in any way.

Solution--

vim lv.yml

```
---
- name: create logical volumes
  hosts: all
  tasks:
    - name: if device not present
      debug:
        msg: device not present
      when: ansible_lvm.vgs.research is not defined

    - name: when vg is present
      community.general.lvol:
        vg: research
        lv: data
        size: 1500m

    - name: when required size is not present
      debug:
        msg: required size is not present
      when: ansible_lvm.vgs.research.free_g>="1.6"

    - name: if not present
      lvol:
        vg: research
        lv: data
        size: 800m
      when: ansible_lvm.vgs.research.free_g>="1.6"

    - name: format lvm
      community.general.filesystem:
        fstype: ext4
        dev: /dev/research/data
```